B-spline Collocation Method

for eigenvalue problems

May 8, 2025

Contents

1 Introduction

In this assignment, we apply the B-spline collocation method to solve the radial Schrödinger equation for the hydrogen atom. We aim to compute the first few state energies and compare them to the known analytical solutions, where different knot sequences are investigated to study the convergence and accuracy of the method.

2 Theory

2.1 Radial Schrödinger Equation

For a hydrogen atom, the radial part of the Schröodinger equation reads:

$$\left[-\frac{1}{2}\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{2r^2} - \frac{1}{r}\right] u(r) = Eu(r), \tag{1}$$

where $u(r) = rR(r)$, $R(r)$ is the radial wavefunction, E is the energy eigenvalue, and ℓ is the angular momentum quantum number.

2.2 B-spline Approximation

The function $u(r)$ is expanded as a linear combination of B-spline basis functions $B_i(r)$ of order k:

$$u(r) = \sum_i c_i B_i(r). \tag{2}$$

The Galerkin method leads to the generalized eigenvalue problem:

$$\mathbf{Hc} = E\mathbf{Sc}, \tag{3}$$

where the Hamiltonian matrix $\mathbf{H}$ and overlap matrix $\mathbf{S}$ are defined by

$$H_{ij} = \frac{1}{2}\int B_i'(r)B_j'(r)\,dr + \int \left[\frac{\ell(\ell+1)}{2r^2} - \frac{1}{r}\right] B_i(r)B_j(r)\,dr, \tag{4}$$

$$S_{ij} = \int B_i(r)B_j(r)\,dr. \tag{5}$$

Integration is carried out using Gauss-Legendre quadrature on each knot interval.

3 Numerical Method

3.1 Knot Sequences

To construct the B-spline basis, we considered three different knot sequences:

- **Linear knots**: Uniformly spaced interior knots in the interval $[r_{\min}, r_{\max}]$:

  $$r_i = r_{\min} + i \cdot \Delta r, \quad \Delta r = \frac{r_{\max} - r_{\min}}{N_{\text{pts}}}, \quad i = 0, 1, \ldots, N_{\text{pts}} - 1.$$

  With k repeated values at each endpoint.

- **Exponential knots**: Clustered near the origin to better resolve behavior close to the nucleus:

  $$r_i = r_{\max}\left(1 - e^{\alpha x_i}\right), \quad x_i \in [0, 1],$$

  $\alpha < 0$ controls the strength of the clustering resulting in more knots near $r = 0$, which is useful for strongly peaked wavefunctions.

- **Quadratic knots**: Also clustered near the origin, but using a polynomial mapping:

  $$r_i = r_{\max} \cdot x_i^2, \quad x_i \in [0, 1],$$

  where x_i are also uniformly spaced.

3.2 Limitations of the Model

Although the B-spline collocation method is powerful, it involves approximations that introduce some limitations:

3.2.1 Physical limitations

Finite wavefunction The real hydrogen atom wavefunctions decay exponentially but extend infinitely, but here we must cut the domain at a finite value $r_{\max}$.

3.2.2 Numerical limitations

Knot points The choice of knot sequence affects the convergence and accuracy. Linear knots may poorly capture the rapid variations near the nucleus, while exponential/quadratic knots offer better resolution.

Finite basis size Using a limited number of B-splines restricts the accuracy achievable for higher excited states.

4 Results

4.1 Hydrogen

4.1.1 Energy levels

We solved the radial Schrödinger equation for the hydrogen atom (with $Z = 1$ and $\ell = 0$) using B-spline basis functions of order $k = 7$, with $N = 20$ basis functions.
The numerical eigenvalues are compared to the known analytical energy levels $E_n = -\frac{1}{2n^2}$ for the first five states as shown in Table 1:

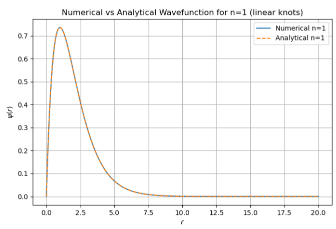
Table 1: Comparison between numerical and analytical eigenvalues for $n = 1$ to $n = 5$.

n	Analytical E_n	Linear	Exponential	Quadratic
1	-0.500	-0.500	-0.500	-0.500
2	-0.125	-0.125	-0.125	-0.125
3	-0.0556	-0.0554	-0.0556	-0.556
4	-0.0312	-0.0309	-0.0312	-0.313
5	-0.0200	-0.0196	-0.0201	-0.0203

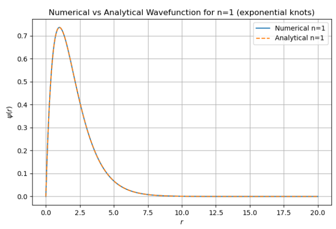
All knot choices agree well with the analytical results, especially for the lowest eigenstates. Small deviations appear for higher states especially for the linear knot sequence due to the poor resolution at smaller r.

4.1.2 Wavefunctions

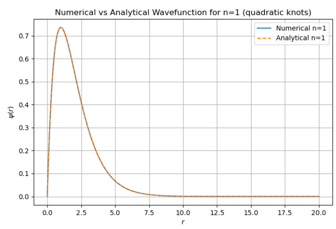
Now we plot the numerically computed wavefunction for the ground state along with the exact analytical solution for all three knot types as shown in in Figure 1:



(a) Linear knots



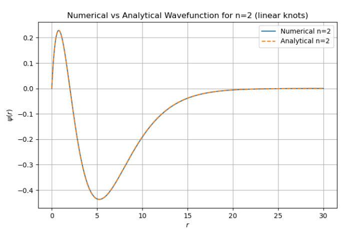
(b) Exponential knots



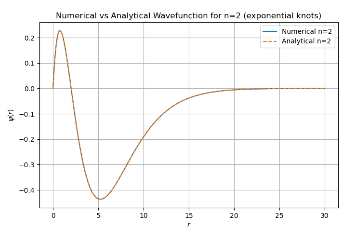
(c) Quadratic knots

Figure 1: Numerical vs Analytical Wavefunction for the ground state $n = 1$ using different knot sequences.

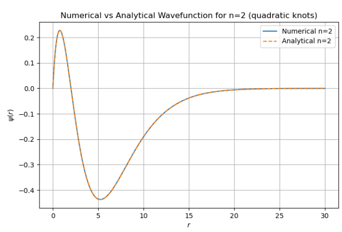
As seen in the figure, the numerical solution matches the analytical one extremely well in all cases, capturing both the exponential decay and the peak near $r = 0.7$.
Figure 2 and 3 shows the wavefunctions corresponding to $n = 2$ and $n = 3$ respectively:



(a) Linear knots

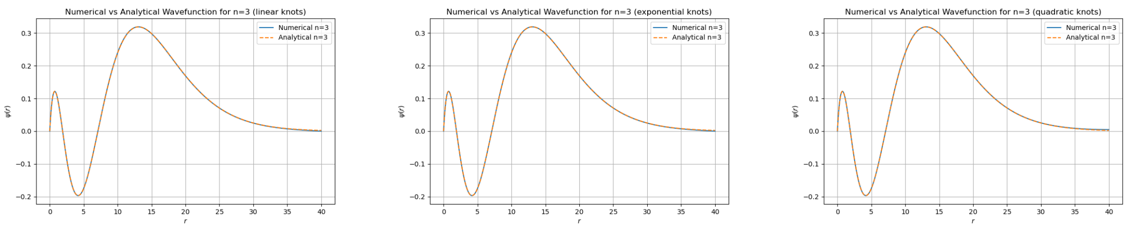


(b) Exponential knots



(c) Quadratic knots

Figure 2: Radial wavefunction for $n = 2$.



(a) Linear knots (b) Exponential knots (c) Quadratic knots

Figure 3: Radial wavefunction for $n = 3$.

As shown in the figures, the computed wavefunctions show excellent agreement with analytical solutions for all knot types. The choice of knot sequence has more impact on higher-energy states, as seen in Table 1.
For lower energy states , all knot types give eigenvalues and eigenfunctions that match the analytical ones. However, as n increases, the eigenvalues computed using linear knots tend to deviate more, since higher-energy states have more than one peak near the origin and extend further out in r, which requires better resolution in both the inner and outer regions.

4.1.3 Eigenvalue Spectrum for different ℓ values

Now we investigate the impact of different angular momentum on the bound state spectrum of hydrogen by computing the eigenvalues (using a linear knot sequence) for $\ell = 1$ and $\ell = 2$.
However, for each value of ℓ, the first allowed bound state corresponds to the state with he smallest quantum number:

$$n \geq \ell + 1 \tag{6}$$

since lower n values are not allowed.

State	**Numerical**	**Analytical**
$n = 2$	-0.124	-0.125
$n = 3$	-0.0553	-0.0556
$n = 4$	-0.0311	-0.0313
$n = 5$	-0.0199	-0.0200
$n = 6$	-0.0125	-0.0139

Table 2: Numerical and analytical eigenvalues for $\ell = 1$.

State	**Numerical**	**Analytical**
$n = 3$	-0.0552	-0.0556
$n = 4$	-0.0312	-0.0313
$n = 5$	-0.0199	-0.0200
$n = 6$	-0.0128	-0.0139
$n = 7$	-0.0045	-0.0102

Table 3: Numerical and analytical eigenvalues for $\ell = 2$.

The numerical eigenvalues show excellent agreement with analytical predictions, particularly for lower energy levels. The small deviations for higher n are expected due to the limitations discussed in 3.2.

4.2 Coulomb vs. Spherical Charge Distribution

We now replace the point-like Coulomb potential with a spherical charge distribution:

$$V(r) = \begin{cases} -\dfrac{Z}{2R_0}\left(3 - \dfrac{r^2}{R_0^2}\right), & r < R_0 \\ -\dfrac{Z}{r}, & r \geq R_0 \end{cases} \tag{7}$$

4.2.1 Hydrogen-like Uranium

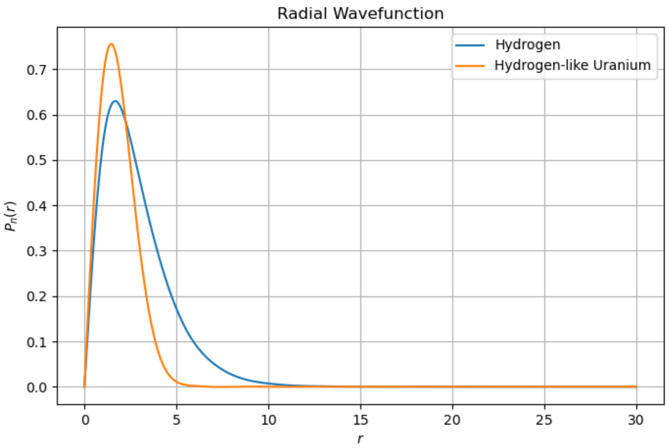
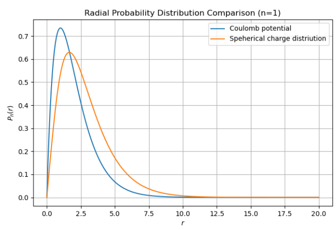


Figure 4: Radial wave function for the ground state of Hydrogen and Hydrogen-like Uranium.

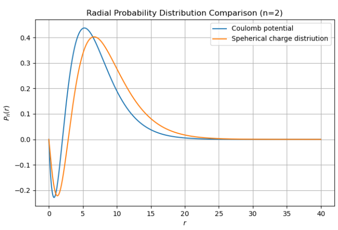
This plot shows the qualitative difference introduced when modeling the uranium nucleus as a finite-sized spherical charge distribution. Compared to hydrogen, the uranium wavefunction is more localized due to the larger nuclear charge ($Z = 92$), that is, the electron is slightly shifted inward with a higher probability to find it near the nucleus relative to the hydrogen case due to the strong attraction force.

4.2.2 Many-electron systems

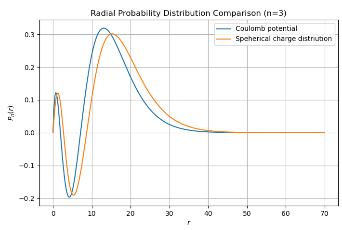
The Figures below show the wave functions (with $R_0 = 1.5\, a_0$ and $Z = 1$) for both the Coulomb and spherical charge potentials.



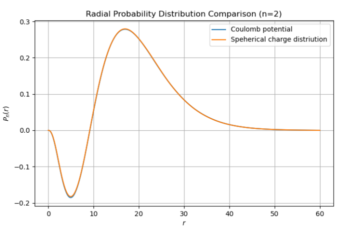
$n = 1, \ell = 0$



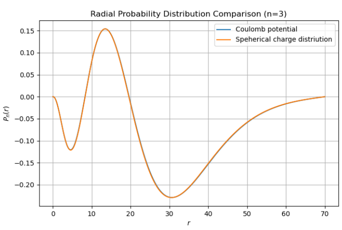
$n = 2, \ell = 0$



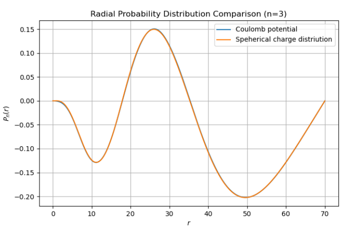
$n = 3, \ell = 0$



$n = 2, \ell = 1$



$n = 3, \ell = 1$



$n = 3, \ell = 2$

The spherical charge distribution for low angular momentum, $\ell = 0$, significantly affects the radial probability distribution near the origin as seen in the plots for $n = 1, 2, 3$, $\ell = 0$, where they are slightly outward shifted compared to the Coulomb case, since the spherical charge potential is less attractive near the origin and hence pushes the electron density further from the nucleus.
However, as ℓ increases, the centrifugal barrier $\ell(\ell+1)/r^2$ dominates and the difference between the two potentials significantly decreases.
In conclusion, using a point-charge potential can be a good approximation unless we are dealing with deeply bound inner electrons (e.g., $1s$-states).

5 Discussion

The B-spline collocation method proved to be effective in solving hydrogen and hydrogen-like problems. The accuracy depends on the knot distribution, where exponentially/quadratically clustered knots provided better results compared to linear knots by better resolving the behavior near the nucleus.